

The Fertility, Marriage, and Gender Equality Quandary

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Motivation

- Three trends underpin the grand gender convergence in the last century:
 1. [Falling fertility](#) (Guinnane 2011)
 2. [Declining marriage](#) (Stevenson and Wolfers 2007)
 3. [Converging gender \(income\) gaps](#) (Goldin 2014)
- Many policies targeting individual outcomes
- Existing research often treats them in isolation or pairwise, lacking a unified framework

This Paper

1. Document a **three-way trade-off**: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically unlikely
⇒ deep dependency structure & potential policy boundaries
2. Develop a **unified model** of marriage, fertility, and female labor supply to explain the empirical facts and study policy implications
3. Calibrate to Mexico (1990–2015), conduct decomposition and prediction

Key Findings

1. Mutual information analysis reveals **synergy**: the three outcomes should be analyzed jointly
2. Traditional policies cannot resolve the trade-off
3. **Redistributing childcare** to husbands can under two conditions
4. Quantitative results on Mexico from 1990 to 2015:
 - Gender-neutral TFP explains half of declining fertility and marriage
 - Gender-biased TFP and education gap reversal drive income convergence
5. Dynamic propagation through gendered impacts of single parenthood

Literature

- Pairwise studies of fertility, marriage, and gender gaps (Galor & Weil 1996, Regalia & Ríos-Rull 2001, Greenwood et al. 2016)
→ Establish three-way trade-off; unified framework
- Gender-biased technical change (Greenwood et al. 2005, Ngai & Petrongolo 2017)
→ Show gender-neutral TFP also drives convergence
- Boys' disadvantage in single-parent families (Autor et al. 2019)
→ Embed in macro model with dynamic propagation
- Policy evaluation in family economics (Doepke & Kindermann 2019)
→ Identify policy frontier; childcare redistribution as remedy

Roadmap

- Empirical findings
- Model
- Policy analysis
- Quantitative analysis
- Dynamic extension
- Conclusion

Motivating Facts

Empirical Approach

1. Create datasets with fertility, dual parenthood, and gender income gap
2. Create dummy variables at sample medians:
 - High fertility (F)
 - Dual parenthood (M)
 - Gender income equality (G)
3. Visualize intersections using Venn diagrams
4. Compute mutual information to characterize dependency structure

Dartboard approach: random benchmark is $0.5 \times 0.5 \times 0.5 = 12.5\%$

OECD Sample

37 OECD countries, 1970–2014

721 observations

- Total fertility rate (1.69)
- Out-of-wedlock birth share (31.4%)
- Gender median earnings gap (17.2%)

Advantage: longer horizon

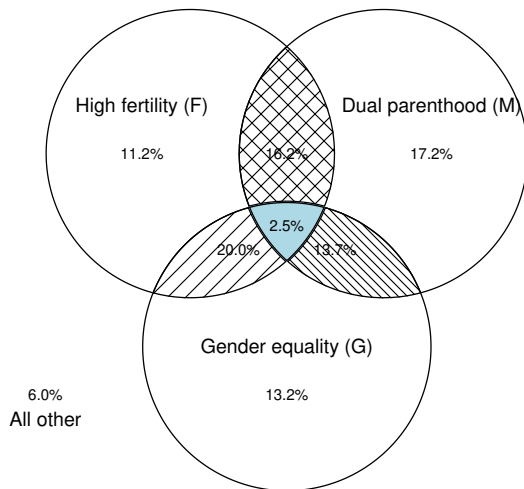
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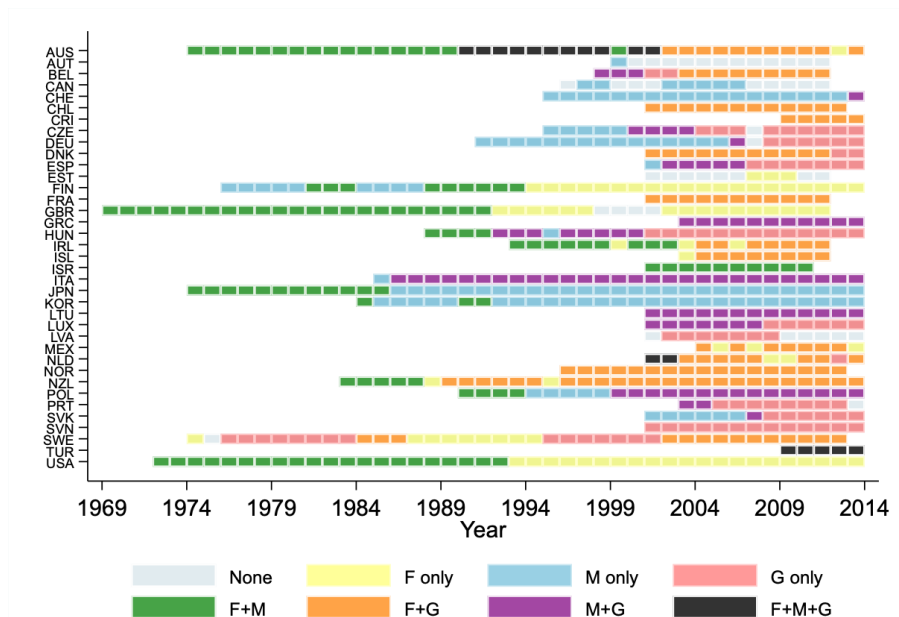
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Trinity share: 2.5%

MI analysis: synergy

OECD Sample – Transition Paths



World Sample

95 countries, 1990–2015

1,130 observations

- Total fertility rate (2.81)
- Children with both parents (75.9%)
- Female labor income share (32.3%)

Advantage: better measurement, wider coverage

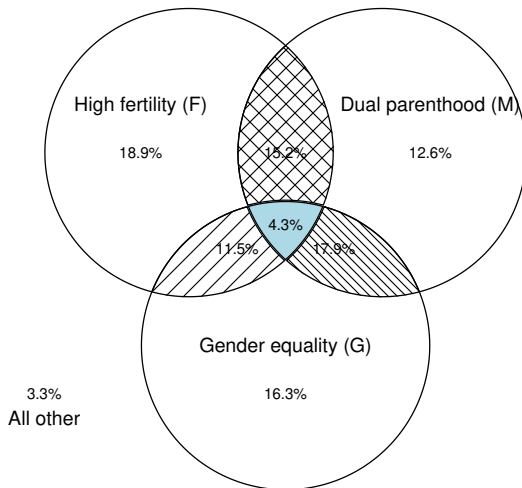
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Trinity share: 4.3%

MI analysis: synergy

U.S. Sample (States)

50 states + D.C., 2023

- Births per 1000 women
(56.9)
- Single-mother household
share (17%)
- Gender earnings gap
(25.6%)

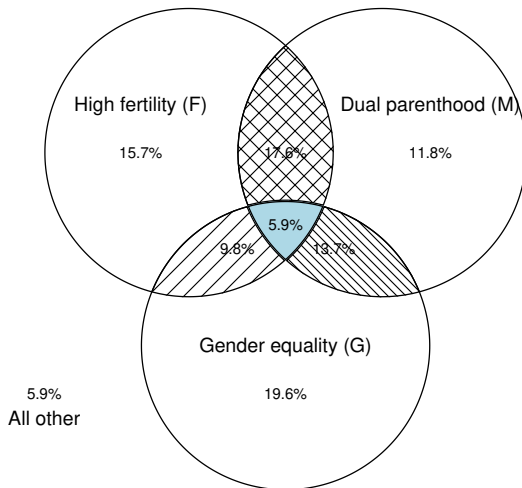
Advantage: more uniform institutional setting

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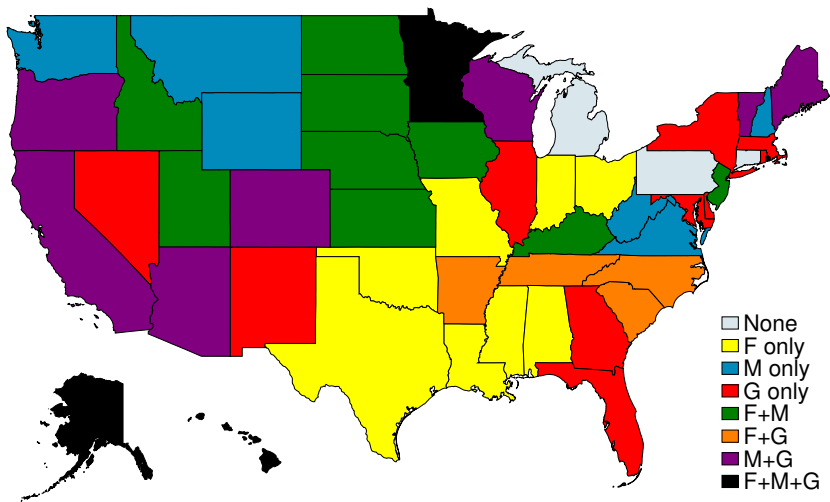
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Trinity share: 5.9%

MI analysis: synergy

U.S. Sample (States) – Geographic Distribution



Questions

1. What explains the three-way trade-off?
2. Are there policies that can help escape the trade-off?
3. What role does technological progress play?

Theory

Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender $g \in \{\sigma, \varphi\}$ and preference

$$u(c, n) = \left((1 - \beta) \cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (1)$$

where $\rho > 1$ so that $u(c, 0)$ is well-defined

- Homogeneous wage within gender: w^σ and w^φ
- Gender wage gap: $\Gamma^w = w^\sigma / w^\varphi$

Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{\sigma^{\rightarrow},s} = u(w^{\sigma^{\rightarrow}}, 0) \quad (2)$$

- Once married, husbands transfer α share of income to wives

$$V^{\sigma^{\rightarrow},m} = u((1 - \alpha)w^{\sigma^{\rightarrow}}, n^m) \cdot \lambda \quad (3)$$

- α is endogenous in marriage market equilibrium
- λ is the psychic benefit of marriage

Marriage and Fertility – Women

- Single women solve

$$V^{\text{♀},s} = \max_{c,l,n} u(c, n) \quad \text{s.t.} \quad c = w^{\text{♀}}l, \quad l = 1 - \chi n \quad (4)$$

- Married women solve

$$V^{\text{♀},m} = \max_{c,l,n} u(c, n) \cdot \lambda \quad \text{s.t.} \quad c = \alpha w^{\text{♂}} + w^{\text{♀}}l, \quad l = 1 - \chi n \quad (5)$$

- Trade-off between consumption and fertility through female labor supply
- In the baseline model, men bear no childcare cost. Results hold as long as $\chi^{\text{♂}} > \chi^{\text{♀}}$

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with scale θ
- Marriage rates:

$$\mathcal{M}^{\sigma} = \frac{1}{1 + (V^{\sigma,s}/V^{\sigma,m})^{\theta}}, \quad \mathcal{M}^{\varphi} = \frac{1}{1 + (V^{\varphi,s}/V^{\varphi,m})^{\theta}} \quad (6)$$

- Equilibrium: $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$
- (α, n^m) act as market-clearing “prices”

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- (α, n^m) act as market-clearing “prices”
- Lemma 1: For any wage pair $\{w^{\sigma}, w^{\varphi}\}$, there exists a unique equilibrium (α, n^m) – intersection between $\alpha(n^m)$ (men) and $n^m(\alpha)$ (women)
- Lemma 2: Gains from marriage: $V^{\varphi,m}/V^{\varphi,s} = \lambda \cdot (1 + \alpha\Gamma^w)$

Three-Way Trade-Off: Intuition

- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha\Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha\Gamma^w))^\theta} \quad (7)$$

$$\Gamma^y = \Gamma^w / l^{\varnothing} \quad (8)$$

$$l^{\varnothing} = 1 - \chi n \quad (9)$$

- Insight:** Higher n reduces $l^{\varnothing}$, therefore needs low Γ^w to achieve gender income equality, but low Γ^w reduces $\mathcal{M}$

Traditional Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\text{♀}}$)	−	−	− (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

- None of these policies can simultaneously improve all three outcomes
- Women's disproportionate childcare burden ties fertility to gender income gaps through relative labor supply

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- None of these policies can simultaneously improve all three outcomes
- Women's disproportionate childcare burden ties fertility to gender income gaps through relative labor supply
- What if the allocation of childcare burden is inefficient?

Redistributing Childcare to Husbands

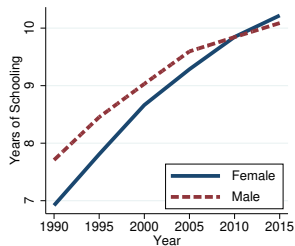
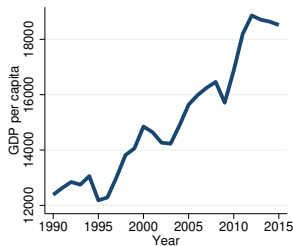
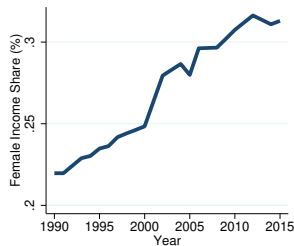
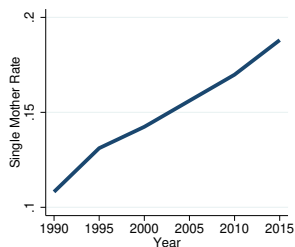
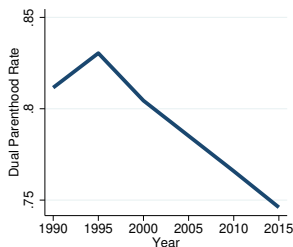
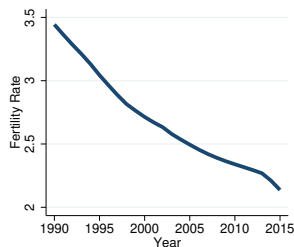
- Extension: $\chi^{\text{♀}}$ for wives, $\chi^{\text{♂}}$ for husbands, $\chi = \chi^{\text{♀}} + \chi^{\text{♂}}$ for single mothers
- Wife's effective childcare cost: $\tilde{\chi}^{\text{♀}} = \frac{\alpha w^{\text{♂}} \chi^{\text{♂}} + w^{\text{♀}} \chi^{\text{♀}}}{\alpha w^{\text{♂}} + w^{\text{♀}}}$
- Consider redistributing δ childcare cost from wives to husbands

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- Consider redistributing δ childcare cost from wives to husbands
- **Proposition:** Redistribution improves all three outcomes if:
 1. Wife's earnings exceed spousal transfers: $w^{\text{♀}} > \alpha \cdot w^{\text{♂}}$
 2. Husband's gain from fertility exceeds consumption loss. More likely if
 - Preference weight on child β is large
 - Initial $\chi^{\text{♂}}$ is small
 - Elasticity of substitution ρ is low
 - Fertility response is strong
- When both conditions hold, redistribution is **Pareto-improving**

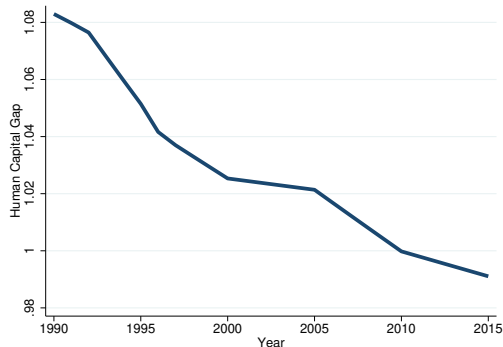
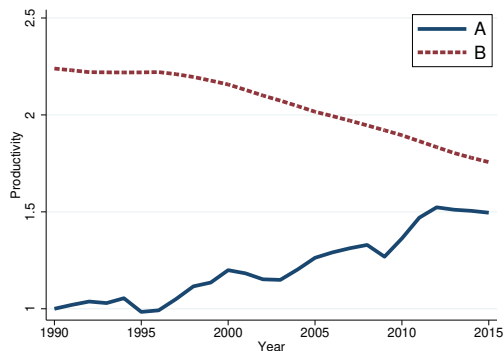
Quantitative Analysis

Mexico's Transition (1990–2015)



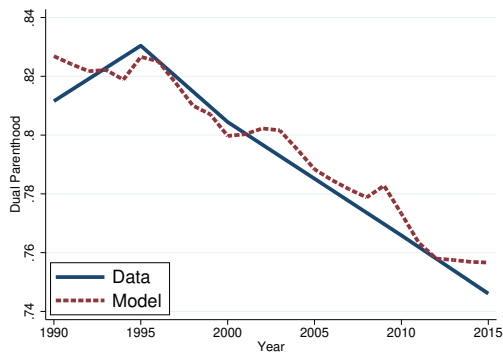
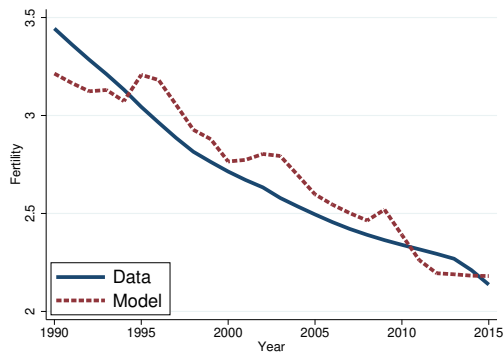
Calibration

- Decompose wages into three components:
 - $A_t = w_t^{\sigma}$: gender-neutral productivity
 - Γ_t^h : gender gap in human capital
 - $B_t = (w_t^{\sigma} / w_t^{\varphi}) / \Gamma_t^h$: gender-biased productivity



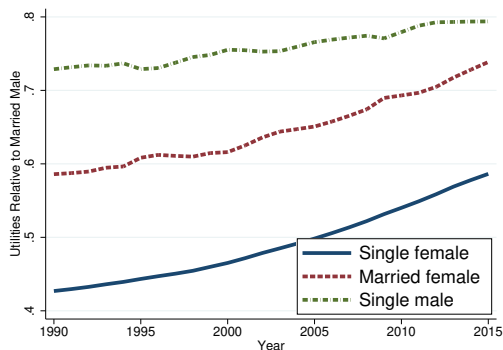
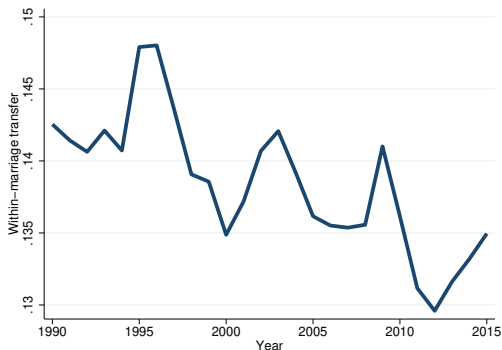
Model Fit

- Fix $\chi = 0.075$ (de La Croix and Doepke 2003) assuming $\chi^{\sigma} \approx 0$
- Calibrate: $\beta = 0.104$, $\rho = 1.5$, $\lambda = 1.03$, $\theta = 3.4$

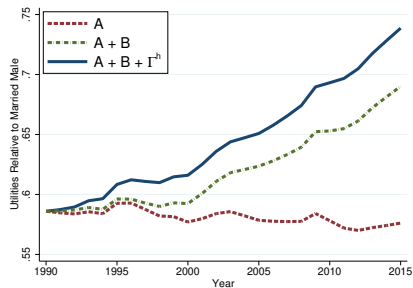
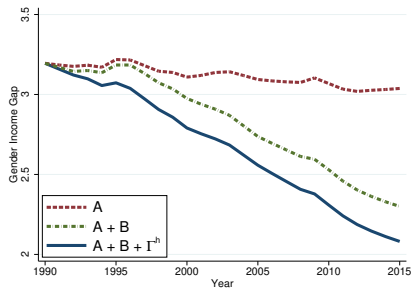
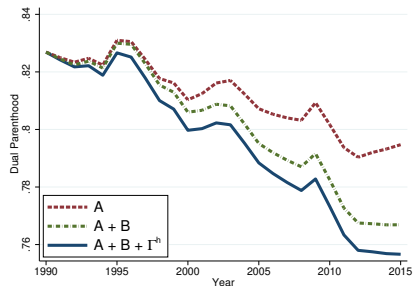
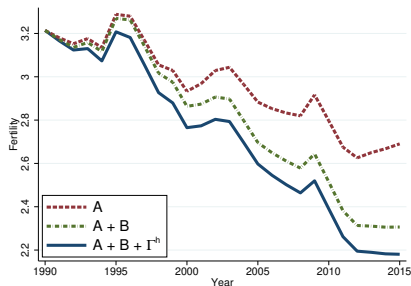


Equilibrium Outcomes

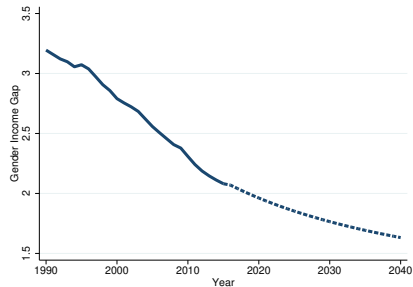
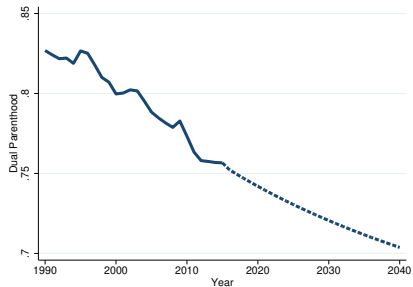
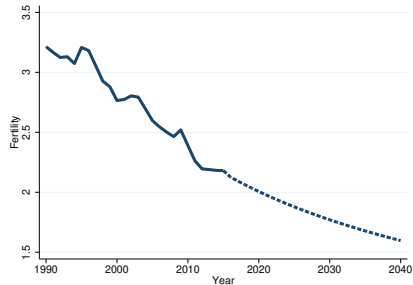
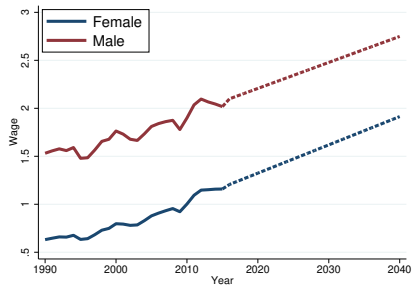
- Small reduction in α despite large drop in n^m as $w^{\text{♀}}/w^{\text{♂}}$ rises
- $\alpha \approx 0.14$ satisfies condition for wives to benefit; husband condition also satisfied $\implies$ redistribution would be Pareto-improving



Decomposition



Model-Based Prediction



Dynamic Extension

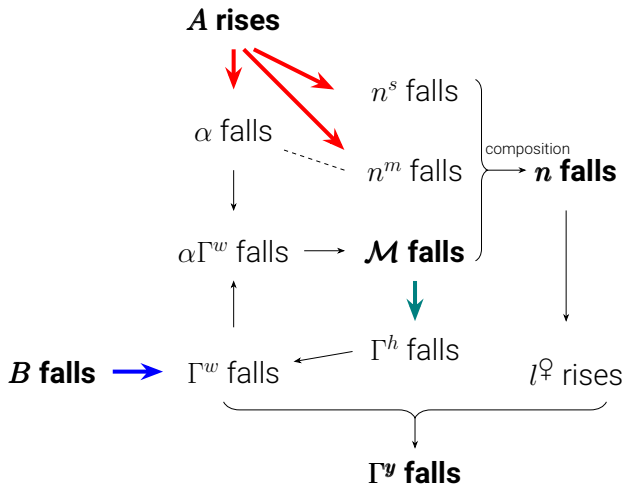
Endogenous Gender Human Capital Gap

- Single parenthood affects boys and girls differently
- Evidence: Bertrand & Pan 2013, Autor et al. 2019, Wasserman 2020, Reeves 2022
- Law of motion:

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t) \quad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0 \quad (10)$$

- Creates self-reinforcing feedback: declining marriage $\rightarrow$ lower male human capital $\rightarrow$ further marriage decline

Dynamic Mechanism



Red: A_t rises Blue: B_t falls Teal: Dynamic feedback

Conclusion

- A three-way trade-off with synergistic dependency structure
- Unified framework explaining the empirical pattern
- Traditional policies cannot resolve the tension; redistributing childcare to husbands can under certain conditions
- Quantitative analysis: both gender-neutral and gender-biased technological change matter
- Dynamic propagation through human capital amplifies effects across generations